

## ❏ 1. Modules in JavaScript (Node.js)

### ◆ 1. Modules kya hote hain?

- Ek **module** JavaScript ki file hoti hai jo kisi **function**, **variable**, ya **logic** ko export karti hai.
- Modules ka use code ko **clean**, **reusable**, aur **manageable** banata hai.

### ◆ Why use Modules ?

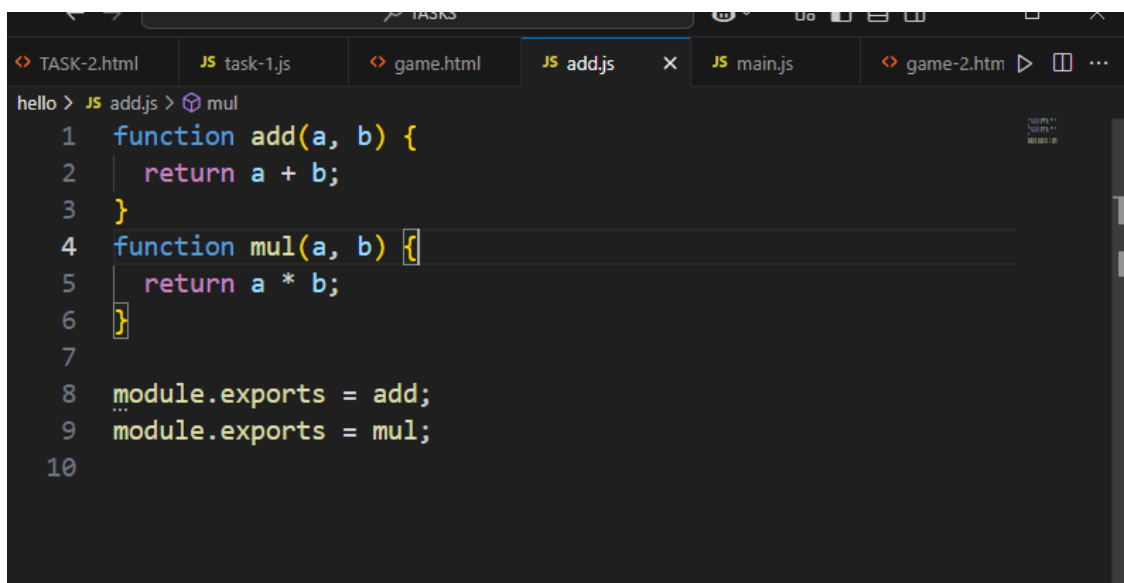
- Reusability
- Better code organization

| Term                        | Description                           |
|-----------------------------|---------------------------------------|
| <code>module.exports</code> | Exports function/variable from a file |
| <code>require()</code>      | Imports a module into another file    |
| CommonJS                    | Module system used in Node.js         |
| <code>add.js</code>         | Custom module file                    |

## Run Karne Ka Tarika (Step-by-step)

1. Dono files (e.g., `functions.js`, `main.js`) ek hi folder me rakho.
2. Terminal khol ke us folder me jao.
3. Command chalao:

```
node main.js
```

A screenshot of a code editor window with a dark theme. The editor has several tabs at the top: 'TASK-2.html', 'JS task-1.js', 'game.html', 'JS add.js' (which is the active tab), 'JS main.js', and 'game-2.htm'. The code in the 'add.js' tab is as follows:

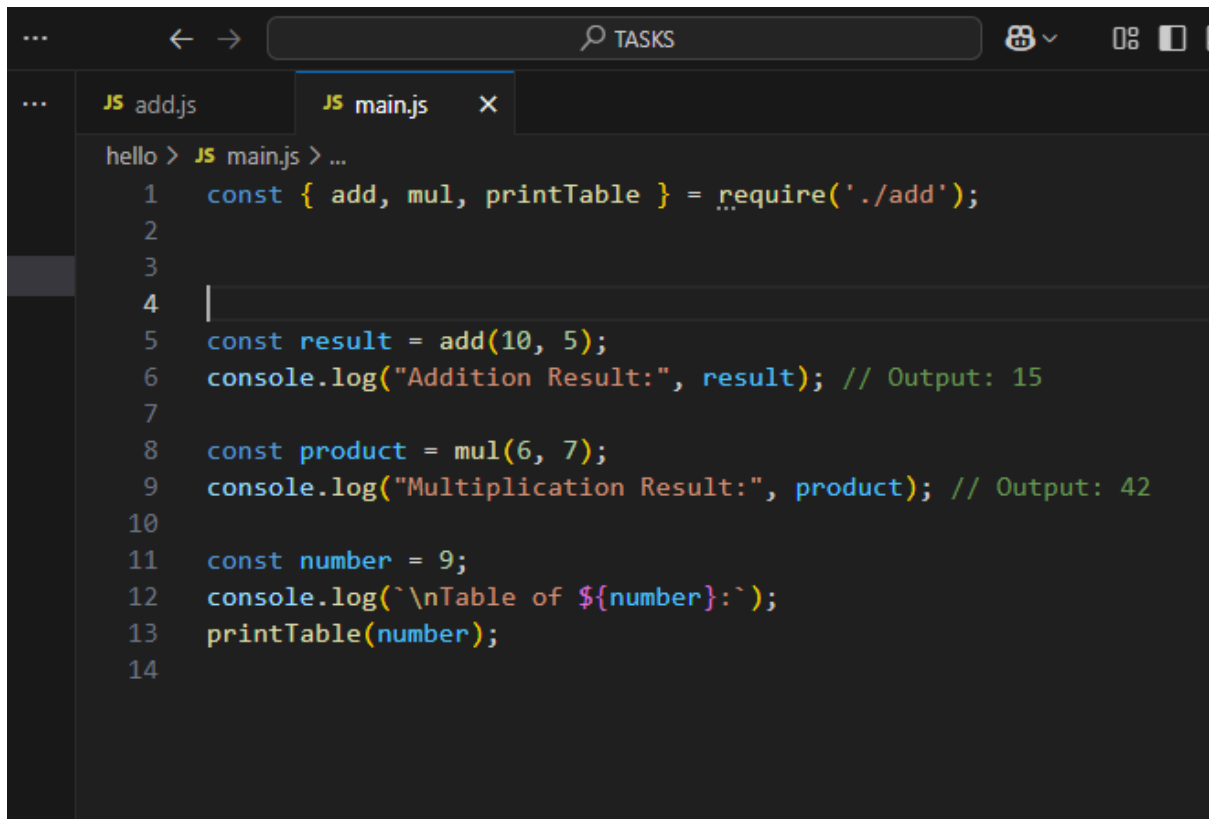
```
hello > JS add.js > mul
1  function add(a, b) {
2      return a + b;
3  }
4  function mul(a, b) {
5      return a * b;
6  }
7
8  module.exports = add;
9  module.exports = mul;
10
```

```
TASK-2.html JS task-1.js game.html JS add.js JS main.js X game-2.htm ▶ □ ...
hello > JS main.js > ...
1 const add = require('./add'); // Importing the function
2 const mul = require('./add'); // Importing the function
3
4 let result = mul(10, 5);|
5 console.log("Result:", result); // Output: Result: 15
6
```

Second Code ➔

```
EXPLORER ... JS add.js X JS main.js
└─ TASKS
  └─ hello
    └─ JS add.js
    └─ JS main.js
    └─ game-2.html
    └─ game.html
    └─ JS script.js
    └─ # style.css
    └─ JS task-1.js
    └─ TASK-2.html

hello > JS add.js > mul
1 function add(a, b) {
2   return a + b;
3 }
4 function mul(a, b) {
5   return a * b;
6 }
7
8 // Function to print multiplication table of a given number
9 function printTable(num) {
10   for (let i = 1; i <= 10; i++) {
11     console.log(`${num} x ${i} = ${num * i}`);
12   }
13 }
14
15 // Export the function
16 module.exports = {
17   add,
18   mul,
19   printTable
20 };
```



```
...    <  →  TASKS  [Icons]
...    JS add.js    JS main.js  X
hello > JS main.js > ...
1    const { add, mul, printTable } = require('./add');
2
3
4    |
5    const result = add(10, 5);
6    console.log("Addition Result:", result); // Output: 15
7
8    const product = mul(6, 7);
9    console.log("Multiplication Result:", product); // Output: 42
10
11    const number = 9;
12    console.log(`\nTable of ${number}:`);
13    printTable(number);
14
```

Execute Only Main.js File

Code ➞

Code

[Click me](#)